Pro Football Sports Science Research Update – January 2025 Audio Summary

Today, we're diving into the article titled "Adductor-Related Groin Injury: Prevalence and Etiology During NCAA Football Games on Artificial and Natural Grass Surfaces," published in PM&R: The Journal of Injury, Function, and Rehabilitation.

This prospective cohort study tracked 41 FBS programs over 16 seasons, analyzing 2,243 games and 9,360 injuries. Out of 111 adductor-related groin cases—making up just 1.2% of total injuries—43% occurred on heavyweight artificial turf and 57% on natural grass. Key finding: No significant differences in injury rates, severity, mechanisms, or situations between surfaces. Most were minor muscle strains from non-contact sprinting during defensive plays, often affecting secondary players.

Bottom line for NFL teams: Heavyweight turf is as safe as grass for groin trauma. Prioritize early detection, structured rehab starting with general lower-body work, and adductor-specific training. Focus on player health over surface debates—this data's NCAA-specific but highly relevant.

Today, we're diving into the 2025 review titled "Advances in Non-Pharmacological Strategies for DOMS: A Scoping and Critical Review of Recent Evidence," published in the Journal of Functional Morphology and Kinesiology.

This scoping review crunches data from 23 high-quality studies between 2020 and 2025 on tackling Delayed Onset Muscle Soreness—that post-eccentric workout pain, stiffness, and performance dip hitting your players hard. Forget pills; it spotlights non-drug options like mechanical therapies (think vibration and percussive massage), thermal like cryosauna, nutritional boosts via polyphenols or BCAAs, and neurophysiological tweaks.

Key takeaway: Multimodal, personalized combos crush single treatments. Standouts include cryosauna at -110°C delayed 48-72 hours post-session to slash stiffness and inflammation markers; 30-40 minutes of percussive massage for better ROM and jump power; and polyphenol supps for faster recovery. Ditch static stretching or electrical stim as go-tos—they underperform. Foam rolling and active recovery? Solid supports, but integrate 'em.

For your gridiron warriors, this means tailoring protocols to cut downtime and optimize performance. Future research calls for standardized measures, but right now, go multimodal to keep your roster firing.

Today, we're breaking down the 2025 article titled "Adverse Effects of Anabolic Androgenic Steroid Abuse in Athletes and Physically Active Individuals: A Systematic Review and Meta-Analysis," published in Substance Use & Misuse.

The authors reviewed 14 studies on AAS abuse—those synthetic testosterone knockoffs chasing muscle gains. Following PRISMA guidelines, they crunched data on risks across cardiovascular, hepatic, endocrine, musculoskeletal, and psychiatric systems.

Key takeaways: AAS ramps up systolic blood pressure by about 12 mmHg and diastolic by 8 mmHg, plus LDL cholesterol by 9 mg/dL—hello, heart disease red flags. Liver enzymes AST and ALT spike by 14 and 12 units, signaling damage. Testosterone crashes by over 140 ng/dL, messing with hormones and fertility. Muscles might hypertrophy, but risks like tendon ruptures loom. Psych-wise, expect aggression, mood swings, and dependency.

Bottom line: For your players, this screams the need for education, routine monitoring, and clean alternatives like optimized training and nutrition. Long-term studies are next—stay vigilant to keep athletes safe and performing legit.

Today, we're breaking down the article titled "Anterior Cruciate Ligament Tear Location in National Football League Athletes Based on Magnetic Resonance Imaging," published in the HSS Journal: The Musculoskeletal Journal of Hospital for Special Surgery.

ACL tears are a nightmare for our players, and this study sheds light on where they happen and why it matters for treatment. Researchers reviewed MRIs from 191 NFL athletes with acute ACL injuries from 2015 to 2019, classified by tear location: proximal avulsion (type I), proximal (II), mid-substance (III), distal (IV), or distal avulsion (V). The big takeaway? Mid-substance type III tears dominated at 74%, with proximal type II at 20%, and only 3% type I—way lower than the 4-26% seen in general populations.

For 73 cases with video analysis, noncontact mechanisms caused 47% of tears, indirect contact 32%, and direct contact 21%. Type III was most common across all, but proximal tears leaned noncontact, while distal ones tied to contact. This suggests higher-energy NFL hits lead to more mid-ligament damage, making preservation techniques like repair less viable—reconstruction remains king.

Bottom line: Tailor your rehab and return-to-play plans knowing most tears are mid-substance. Stay vigilant on prevention drills to cut noncontact risks.

Welcome to this quick dive into a key paper from Sports Medicine journal, published online November 27, 2025. Titled "Assessing Deceleration Performance: Methodological and Practical Considerations," by Damian Harper and colleagues.

In multi-directional sports like football, deceleration is a game-changer—letting players slam on the brakes for quick cuts while handling massive forces that spike injury risks like ACL tears or fatigue. But we've historically fixated on acceleration and top speed, ignoring how to stop safely from those high velocities.

This article flips the script, urging us to prioritize decel assessments. It breaks down two main protocols: Acceleration-Deceleration Ability (ADA) tests, like sprinting 10-20 meters then stopping sharp, and Change of Direction (COD) tests with 90-180 degree turns, such as the 505. Key is standardizing approach speeds and ensuring velocity hits zero for accurate metrics like average decel in m/s^2, distance, or time to stop.

Tech options include radar or laser for instant velocity tracking, motorized resistance devices for high-freq data, plus IMUs, GNSS, or video. Pros: Reliable for field use; cons: Some limit multi-plane moves.

Bottom line: Integrate decel testing into your batteries to boost performance, rehab, and injury prevention—tailor by position, from low-speed rehab to high-momentum game sims. Don't sleep on this; it could save seasons.

Here's a quick research spotlight on foot health in football players. The study, titled "Assessment of Static Foot Posture as an Indicator of Biomechanical Adaptation in American Football Players," was published in Acta of Bioengineering and Biomechanics.

They compared 70 male Polish American football athletes—averaging 22 years old, with varying training experience—to 35 non-athletes, using the Foot Posture Index-6 to evaluate static foot shape, plus BMI and age factors.

Results? Both groups mostly had neutral feet, but players showed significantly higher BMI (overweight on average) and more pronation tendencies—up to four times higher than controls in some measures. No strong ties between BMI, age, or training years and foot posture in athletes, but controls linked higher BMI to left-foot changes.

Bottom line: Pronation from the sport's demands could raise injury risks like ankle sprains or knee issues. Pro tip—regularly screen foot posture and BMI, and consider custom anti-pronation cleats or insoles to optimize performance and prevent downtime.

Let's dive into a fresh 2025 review on optimizing ACL recovery. The article, "Blood Flow Restriction Training Prior to and After Anterior Cruciate Ligament Reconstruction: A Scoping Review," published in the Journal of Functional Morphology and Kinesiology.

ACL tears hit hard in football, causing muscle atrophy and slow returns to play. Blood flow restriction (BFR) training—using cuffs to limit blood flow during low-load exercises—emerges as a game-changer. Pre-surgery, it boosts quad size and strength; post-op, it matches or beats traditional rehab in muscle gains, power, function, and even biomarkers reducing atrophy.

Key takeaway: BFR allows effective training without heavy loads, speeding rehab while cutting risks like pain or swelling. For your athletes, integrate it early to slash downtime and enhance return-to-play metrics. More RCTs needed, but this could level up your protocols.

Welcome to this quick dive into a key paper: "Calf Strains in Athletes: A Narrative Review of Management, Injury Grading, and Return to Sport," published in Sports Medicine - Open.

Calf strains hit hard in high-demand sports like football, with high recurrence rates and no universal guidelines. This review synthesizes 11 studies to spotlight how imaging like ultrasound and MRI, plus grading systems, predict return to play.

Key takeaways: MRI and US reveal injury extent—think soleus dominance in re-injuries and aponeurotic disruptions adding 8-12 days to recovery. Grading correlates with RTP timelines: mild strains back in 2-3 weeks, severe up to 12. But clinical assessments remain king—pain-free walking, max-speed running, and full training.

For you in the league, note NFL-specific data: Acute gastroc-soleus tears average 17 days out, with larger fascial defects signaling over two weeks sidelined.

Bottom line: Blend imaging with clinical evals for smarter protocols, cutting re-injury risks. Standardize grading for better outcomes—your players will thank you.

Today, we're diving into a key study: "Does strengthening the cervical spine musculature enhance neck strength and reduce sports-related concussions in rugby and football players? A systematic literature review," published in the Journal of Bodywork & Movement Therapies.

With rising concerns over concussions in contact sports, this review analyzed 10 high-quality studies involving over 3,400 athletes—mostly rugby players—from 2002 to 2022. Researchers focused on randomized trials and cohorts testing cervical muscle strengthening programs, like isometric holds and resistance exercises.

Key takeaways: These programs cut concussion rates by nearly half in male rugby players, with a risk ratio of 0.55. They also boosted neck strength, especially in extension. But evidence is slim for soccer players or female athletes—no clear benefits there yet.

For NFL teams, this suggests incorporating neck strengthening into training could help mitigate head impacts, similar to rugby's high-contact demands. More research is needed, but it's a promising tool for player safety.

Today, we're diving into a study titled "Effect of a 90-Minute Nap at Different Times of the Day on Physical Performance, Psycho-Cognitive Responses, and Perceived Recovery in Trained Youth Male Athletes," published in the journal *Sports*.

Researchers examined 14 trained youth male athletes, average age 18, using a randomized crossover design. Participants tested four conditions: no nap or a 90-minute nap starting at 1 PM or 3 PM, followed by a 5-meter shuttle run test to measure sprint performance, fatigue, perceived exertion, recovery, sleepiness, and cognitive attention.

Key findings: A 1 PM nap significantly boosted total and best distance covered, improved attention via digit cancellation tests, and enhanced perceived recovery 24 hours later compared to a 3 PM nap or no nap. Athletes felt sleepier before the later nap but more alert after the early one. No major changes in fatigue index or mood.

For NFL teams, this suggests scheduling 90-minute naps around 1 PM could optimize afternoon practices or games, aiding recovery and repeated sprint ability—crucial for positions like receivers or defensive backs. Consider individual factors like sleep debt.

Today, we're diving into a fresh systematic review and meta-analysis titled "Effects of Resisted-Sprint Training on Sprint Performance and Mechanics: A Systematic Review and Meta-Analysis Focusing on Load Magnitude," published in the Scandinavian Journal of Medicine & Science in Sports.

This study crunched data from 49 resisted sprint training (RST) trials involving over 1,200 athletes, mostly males. Key takeaway: RST edges out unresisted sprint training (UST) for boosting acceleration, with a moderate effect size. But load matters—moderate (20-50% body mass) and very heavy loads (80%+ body mass) outperform UST, while light loads (<20% BM) are on par. Very heavy RST shines for trained pros like your players, ramping up horizontal force, max power, and force ratio—crucial for those explosive 0-10 yard bursts in football.

Effects amp up beyond 20.7% body mass, but watch volume: an inverted U-shape means optimal gains at balanced doses. Factors like age, sex, and sessions tweak results. Bottom line: For NFL speed drills, go heavier on sleds for elite athletes to maximize early acceleration without overdoing volume.

Today, we're breaking down a fresh study from BMC Sports Science, Medicine and Rehabilitation: "Electromyography Activity of Posterior Oblique Sling Muscles During High-Speed Running in Elite Soccer Players: Risk Factors for Hamstring Injury.

Hamstring strains are a nightmare in high-speed sports like football, and this research spotlights how activation patterns in the posterior oblique sling—or POS—muscles (glute max, hamstrings, and contralateral lumbar erectors) could flag injury risks. Researchers tested 24 elite male soccer players, excluding those with ROM issues, and grouped them based on the Prone Hip Extension test: non-delayed activation (NDG) versus delayed glute and hamstring firing (DG).

During 20 km/h treadmill sprints, EMG showed key differences: NDG had higher glute max early in the stride (7-10%), potentially stabilizing the pelvis and reducing hamstring load. DG showed elevated hamstring activity mid-cycle (45-70% for semitendinosus, varying for biceps femoris), plus imbalanced ratios like higher biceps-to-glute during stance—suggesting overload and fatigue risk.

Bottom line: Delayed POS patterns in low-load tests may persist in sprints, hiking hamstring strain odds. For your athletes, consider PHE screening and targeted motor control drills to optimize sling synergy and keep those hamstrings intact.

Today, we're breaking down the systematic review titled "Emerging Use of BPC-157 in Orthopaedic Sports Medicine," published in the HSS Journal: The Musculoskeletal Journal of Hospital for Special Surgery.

BPC-157 is a naturally occurring gastric peptide that's gaining buzz for its potential in healing musculoskeletal injuries like fractures, tendon ruptures, ligament tears, and muscle strains. This review synthesizes 36 studies—mostly preclinical animal models, with just one small human trial—from 1993 to 2024.

Key takeaways: In labs and animals, BPC-157 boosts growth hormone receptor expression, ramps up cell growth, angiogenesis, and nitric oxide pathways while dialing down inflammation via reduced cytokines like IL-6 and TNF-alpha. It showed improved functional, structural, and biomechanical recovery in muscle, tendon, ligament, and bone injuries. That one human study?

Seven out of 12 patients with chronic knee pain reported relief lasting over six months after an intra-articular injection.

On the flip side, it's metabolized in the liver with a half-life under 30 minutes and cleared by the kidneys—no adverse effects in preclinical safety checks across organs. But here's the red flag: It's not FDA-approved, banned by the NFL and WADA, and unregulated products could be contaminated or risky due to zero clinical safety data in humans.

Bottom line for you in the league: Counsel players on the bans and unknowns—stick to compliant options for recovery.

Today, we're unpacking the review article titled "Genetic Susceptibility to Sport-Related Muscle Injuries: Insights from the Literature and Novel Gene Candidates," published in the International Journal of Molecular Sciences.

Physical activity boosts health, but it ramps up muscle injury risks—like those nagging hamstrings that sideline your players. This review synthesizes data from PubMed, Scopus, and Web of Science, spotlighting 37 single nucleotide polymorphisms in 32 genes tied to injury risk, severity, recurrence, and recovery time. These genes fall into six buckets: vascular function (like ACE for blood flow), muscle structure and ECM (think ACTN3's R577X variant, where the XX genotype hikes non-contact injury odds in footballers), metabolism (AMPD1 affecting energy regen and fatigue), inflammation (IL6 influencing post-exercise soreness), growth and regeneration (IGF2 for repair), and membrane signaling.

Key takeaway? Genetics explain why some athletes bounce back fast while others linger on IR. For your teams, this points to personalized screening—genotype players to tailor warm-ups, training loads, and rehab protocols, potentially slashing those €365K monthly costs seen in pro soccer.

Gaps remain, but novel candidates like SOX15 and HGF offer fresh targets. Time to integrate genomics into your playbook for peak performance and fewer DNPs.

Here is a quick hit on the latest research from the Journal of Head Trauma Rehabilitation. The article, titled "Higher Cardiovascular Risk Factor Burden Associated With Cumulative Concussion History and Age-Related Differences," dives into how repeated concussions might accelerate heart health issues in former pros.

Researchers surveyed nearly 2,000 ex-NFL players, splitting them into under-50 and over-50 groups. Key finding: Younger retirees with more concussions—especially 6 or more—showed

significantly higher rates of hypertension and high cholesterol, with a dose-response effect. Sleep apnea was linked to concussion history across all ages. These risks? They correlated with worse depression, anxiety, cognition, and behavioral issues in the under-50 crowd.

Bottom line for us: Concussions could fast-track CV problems, upping long-term neuro risks. Prioritize early screening and management for high-concussion players to protect their futures.

Today, we're breaking down a 2025 study from the Journal of Strength and Conditioning Research titled "Interlimb Asymmetry Data in Athletes With Anterior Cruciate Ligament Reconstruction: A Comparison of Different Equations to Interpret Between-Limb Difference Data".

This research looked at 53 amateur athletes, about 28 months post-ACL reconstruction, testing strength and jumps like 5RM rear-foot elevated split squats, leg extensions, single-leg countermovement jumps, and drop jumps. The big question: How do different equations for calculating interlimb asymmetries affect results during rehab and return-to-sport decisions?

Key takeaway: Equations matter—a lot. They found large differences in asymmetry magnitudes between formulas, which could flip a "pass" to a "fail" on that classic 10% threshold. For unilateral tasks, they recommend the Standard Percentage Difference with direction to capture both size and side. For bilateral tasks like CMJs, go with Bilateral Asymmetry Index 1.

Bottom line for your ACL protocols: Pick consistent equations to avoid misleading data. This could sharpen RTS calls and reduce re-injury risks.

Today's spotlight: "Limited Agreement on ACL Tear Location Between Arthroscopy and MRI—A Prospective Evaluation," published in The Orthopaedic Journal of Sports Medicine.

ACL tears hit over 200,000 Americans yearly, including plenty of our players, with reconstruction as the go-to fix. But with rising interest in preservation and repair for proximal tears, accurate tear location matters big time. This prospective study from Washington University tracked 174 patients with full-thickness tears, excluding partials or multis. Using the modified Sherman classification, MRI showed mostly midsubstance tears—67% type III, just 10% proximal avulsions (type I). But arthroscopy painted a different picture: 26% type I proximal avulsions, 45% type II proximal, and only 26% type III.

Key takeaway: Only 43% agreement between MRI and scope overall, though 96% for midsubstance tears. MRI undercalled proximal tears, which were twice as common on scope—potentially missing repair candidates. Tears skewed proximal in older folks, midsubstance in

younger athletes. No big ties to sex, BMI, or mechanism, but indirect contact hinted at fewer midsubstance rips.

For NFL teams, this means don't bank solely on MRI for tear typing—scope is gold standard. It could shift eval protocols and repair discussions, especially for vets vs. rookies.

Today, we're breaking down the article "Managing Fatigue in Team Sports: A Brief Review of Concurrent Training Effects Within the Microcycle," published in the Strength and Conditioning Journal.

In high-intensity team sports like football, concurrent training—mixing resistance work with energy system conditioning like HIIT or small-sided games—is standard. But it amps up fatigue, risking the interference effect where one session hampers the next, potentially stalling gains in strength, power, or endurance and boosting injury odds.

Key variables to tweak in your microcycle: Dial intensity high for strength (over 80% 1RM) but avoid failure to cut recovery time. Keep volumes moderate—aim for 12 weekly sets per muscle group—to prevent overload. Prioritize session order: Strength before conditioning when possible to minimize molecular clashes, though HIIT might not interfere as much.

Don't forget non-training factors: Optimize sleep, nutrition, and travel recovery. During congested schedules, microdose sessions and monitor with tools like velocity loss or RPE.

Bottom line for the league: This adaptive approach hits the 'sweet spot'—boosting performance while slashing over- or under-training risks. Implement these tweaks to keep your athletes fresh and dominant on game day.

Today, we're breaking down the article titled "Mapping of the Pharmacological Effect of Prohibited Substances and Methods to Athletic Physiological Characteristics," published in *Drug Testing and Analysis*.

In this study, experts created a novel framework to evaluate how WADA-banned substances—like anabolic steroids, EPO, and beta-blockers—affect performance across six key athletic traits: aerobic endurance, muscular endurance, strength and power, reaction time, agility, and psychomotor skill. They rated 31 drug categories and mapped them to 160 Olympic sports, using a scoring system to highlight ergogenic benefits or drawbacks.

Key takeaways? Anabolics and growth hormones shine in power-heavy sports like weightlifting—think NFL linemen building explosive strength. EPO boosts endurance for marathon-like demands, while narcotics and cannabinoids often hinder reaction time and

accuracy, crucial for quarterbacks or receivers. Beta-blockers? Mostly detrimental except in precision scenarios.

For you in NFL sports med and performance: This underscores targeted testing—focus on anabolics for strength gains, but watch for masking agents like diuretics. It optimizes resources, refines risk assessments, and promotes fair play.

Today, we're diving into the article titled "National Football League (NFL) 'Dynamic Kickoff' Rule Change: A Comparative Analysis of Concussion and Musculoskeletal Injury Incidence," published in The Orthopaedic Journal of Sports Medicine.

In 2024, the NFL rolled out the Dynamic Kickoff to cut down on injuries—especially concussions—while boosting returns. This descriptive epidemiology study crunched data from the 2022, 2023, and 2024 regular seasons, pulling injuries from game books and team reports, then calculating incidence rate ratios.

Key takeaways: Overall injuries spiked in 2024 compared to both prior years—up 10% vs 2022 and 59% vs 2023 per kickoff. Upper extremity injuries jumped dramatically, 6.7 times vs 2022, and lower extremity doubled or more. Concussions? Mixed bag—down 23% vs 2022 but up nearly 4 times vs 2023 per kickoff.

Bottom line: The rule boosted returns but didn't curb overall or musculoskeletal risks; concussion impact remains unclear after one year. As pros, keep monitoring adaptations—maybe push for tweaks like expanded touchback zones. More seasons' data needed.

Let's dive into a fresh study from BMC Sports Science, Medicine and Rehabilitation titled "Persistent Neuromuscular Deficits in the Posterior Kinetic Chain Following Hamstring Strain Injury: EMG Insights from Nordic Hamstring Curl, Kettlebell Swing, and Supine Sliding Leg Curl," .

Hamstring strains are a nightmare in football—top non-contact injury, with high recurrence and time lost. This research spotlights lingering neuromuscular issues post-injury that could spike reinjury risk.

Researchers compared 42 male soccer players—21 with a recent unilateral hamstring strain, 21 healthy controls, average age 23. Using surface EMG, they measured activation in the biceps femoris, gluteus maximus, latissimus dorsi, and iliocostalis during three exercises: the knee-dominant Nordic Hamstring Curl and Supine Sliding Leg Curl, plus the hip-dominant Kettlebell Swing.

Key findings? Injured athletes showed significantly lower muscle firing across the board—up to 29% less in biceps femoris and 24% in glutes, with biggest gaps in hip-dominant moves. Even trunk muscles like lats and erectors were down 12-17%, hinting at chain-wide inhibition.

Bottom line: Even after RTP, these deficits persist beyond the hamstrings. Rehab pros, amp up multi-segmental posterior chain work, especially hip-focused drills, to rebuild activation and cut reinjury odds.

Today, we're diving into the 2025 study from the Journal of Strength and Conditioning Research titled "Position-Specific Longitudinal Assessment of Total and Regional Body Composition in NCAA Division I Football Players,".

These researchers tracked 188 Division I college players across nine positions—like DBs, linemen, QBs, and more—using DXA scans from preseason to postseason. They measured total and regional lean mass, fat mass, bone metrics, and muscle-to-bone ratios, controlling for play time and injuries.

Key takeaways: No big shifts in total body mass, fat, or bone across the board. But lean mass dropped significantly in most positions—mainly in the legs and gynoid regions (think hips and upper thighs). This happened across all groups, except QBs and LBs held steady overall. Muscle-to-bone ratios also declined regionally, hinting at potential power and stability losses.

For you in the NFL: This underscores the need for in-season strategies to preserve lower-body lean mass—maybe ramp up targeted resistance training or nutrition tweaks. Use DXA for monitoring to tailor programs and prevent performance dips or injuries.

Today, we're diving into the 2025 article from the *Scandinavian Journal of Medicine & Science in Sports*: "Quadriceps Morphology 6- to 60-Months After Anterior Cruciate Ligament Tear," .

This exploratory cross-sectional study compared 21 young adults median 23 months post-ACL tear—78% reconstructed—against 20 uninjured controls. They assessed bilateral knee extensor strength, rate of torque development, and vastus lateralis biopsies for fiber-type specific cross-sectional area, proportions, and extracellular matrix.

Bottom line: The ACL group showed clear deficits in quad concentric strength (down 39 Nm), eccentric strength (down 30 Nm), and torque development. But surprisingly, no differences in muscle fiber size, type distribution, or fibrosis between groups or legs.

For your athletes, this suggests persistent quad weakness isn't from gross morphology changes—think neural impairments or myofibrillar protein loss instead. Key takeaway: Push

extended resistance training beyond formal rehab, as ongoing knee-focused work (median 2 days/week here) might help normalize muscle. Graft type could play a role too—knee extensor grafts hinted at more atrophy in descriptives.

Today's spotlight: the research article titled "SARMs, Metabolic Modulators and Growth Hormone Secretagogues in Suspected Illegal Medicines, Bought as Sport Performance Enhancers: A Retro- and Prospective Study Within the GEON," published in Drug Testing and Analysis.

The study, from the European OMCL network, analyzed 324 seized samples over five years from 13 countries, focusing on illegal products peddled as muscle builders. Key culprits? SARMs like ligandrol and ostarine, metabolic modulators like cardarine, and growth hormone boosters like ibutamoren—making up the top five detections.

Alarmingly, 65% posed as legit meds, 24% as supplements, fooling users into thinking they're safe. Most came from illegal channels like customs busts or online black markets. Quantitative tests showed active or even overdosed levels, raising red flags for liver damage, heart issues, and other side effects.

Bottom line: These unapproved substances are rampant in recreational sports and could slip into pro circles. Stay vigilant—test rigorously, educate players on risks, and remember, WADA bans them for good reason. Health over gains, always.

Today, we're diving into the article titled "Summary of the 2025 Professionals in Nutrition for Exercise and Sport '10 Questions/10 Experts' Session—Can Everyday Foods Replace Some Ergogenic Supplements and Commercially Available Sports Foods?" published in the International Journal of Sport Nutrition and Exercise Metabolism.

This piece recaps a rapid-fire session at the ACSM Annual Meeting, where 10 experts tackled myths on swapping supplements and sports products with everyday foods. Key takeaways: Dietary creatine intake is often too low from foods alone—busted, consider supplements for optimal health. Nonsugar sweeteners for cutting calories? Equivocal—evidence is mixed, especially for athletes' carb needs. Beef jerky or chocolate milk for leucine? Plausible for women, but men might need BCAAs in deficits. Honey or applesauce as gel replacements? Confirmed—they work just as well for fueling. Homemade juice-salt drinks? Plausible, but match commercial electrolyte profiles. Vegetarians and carnosine/carnitine? Busted—no need unless chasing edges. Teff grain for iron? Plausible, depending on soil contamination. Coconut

water for electrolytes? Busted—add sodium for real benefits. B vitamins skewing urine color hydration checks? And pre-ex calcium on bone metabolism? Stay tuned for full details.

Bottom line: Food-first is smart, but not always sufficient—tailor to your players' demands for peak performance and recovery.

Today, we're diving into a 2025 systematic review and meta-analysis titled "The Correlations Between Training Load Parameters and Physical Performance Adaptations in Team Sports," published in Sports Medicine - Open.

This study crunched data from 40 papers involving 726 athletes—mostly soccer players, but also rugby, basketball, and others—to explore how training loads link to fitness gains over at least two weeks.

Key takeaways: Heart rate-based TRIMP (training impulse) showed strong ties to endurance boosts, like a large correlation (r=0.63) with VO2max improvements, and moderate ones with lactate threshold velocities (r=0.43 to 0.47). Session-RPE had smaller links, such as r=0.29 with lactate thresholds. External loads, like total distance or accelerations, had weak or no correlations with strength, power, or speed adaptations.

For you in the NFL: Prioritize HR-TRIMP for monitoring aerobic progress during training blocks, but don't rely solely on external metrics for neuromuscular gains—more research needed there. This could refine your load management to optimize player adaptations and reduce injury risk.

Today, we're breaking down a 2024 study from *Experimental Physiology* titled "The Impact of Long-Haul Travel and 13-Hour Time Change on Sleep and Rest-Activity Circadian Rhythm in Speed Skaters During World Cup Competitions,".

The researchers tracked 19 elite Canadian short-track speed skaters—average age 24, including Olympians—as they flew from Montreal to Asia, crossing about 13 time zones for World Cup races in 2017 and 2019. Using wrist actigraphs, they monitored sleep, circadian rhythms via rest-activity patterns, neuromuscular jumps, and race times.

Key takeaways: Travel slashed sleep to around 6 hours and 16 minutes versus 7 hours baseline, but during the Asia stay, sleep rebounded to 7.5 hours with better efficiency. It increased by 9 minutes per night over the first 5 days, while circadian activity shifted from evening peaks to mid-afternoon. Jump heights actually improved, and race performances matched their season averages—no jet lag hangover.

For NFL teams jetting coast-to-coast, this suggests athletes can adapt in under a week with smart recovery. Prioritize sleep hygiene post-travel to keep your players sharp.

Welcome to this quick hit on a key paper from Sports Medicine journal. The title? "The Risks of Misunderstanding the Term 'Risk Factor': A Primer with Suggestions to Improve Sports Medicine,"

Look, we throw around "risk factor" all the time—like previous ACL surgery increasing tear risk or poor knee posture hiking lower extremity injuries in football players. But here's the rub: the term's fuzzy. Some mean it causes the outcome; others just see association without causation. Mix that up, and you're chasing wasteful interventions, like tweaking a non-causal marker that won't budge injury rates.

The authors break it down: Associations can be causal (change the factor, change the outcome) or non-causal (due to confounders like shared causes or selection bias). Misunderstandings stem from vague definitions, overusing "modifiable" without causal proof, and our brains' "causal illusion" bias.

Their fix? Ditch ambiguity. Call it a "causal risk factor" if evidence shows intervening helps; "noncausal" if not; or "risk marker" if you're neutral. Define terms in reports, push journals for clarity, and educate teams on causality basics to avoid ineffective programs.

For us in the NFL, this means smarter screening and training—focus resources on proven causes to keep players on the field.

Today, we're spotlighting a 2025 systematic review from *Acta Psychologica* titled "The Role of Personality Traits in Athlete Selection: A Systematic Review".

This PRISMA-guided analysis crunched 22 studies with over 4,000 athletes across 18 sports and 14 countries. Key takeaway: Traits like high conscientiousness and extraversion consistently predict elite performance—with effect sizes from 0.26 to 0.70—while high neuroticism drags it down, linked to poorer emotional stability.

Mechanisms? Think direct brain wiring for reward and focus, plus indirect boosts via motivation and self-regulation, moderated by sport type—like team demands in football favoring extraverts.

For you in the NFL: Integrate tailored personality screens into scouting and development to spot resilient talents early. More longitudinal studies needed, but this could refine your draft picks and injury prevention strategies.